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End-to-End Sequence Modeling of Crop Rotations with Satellite Image Time Series

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Ringraziamenti

...thanks to...

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Abstract

Precision agriculture is increasingly vital for ensuring global food security and sustainable resource management, heavily relying on the accurate, large-scale mapping of land use and crop distribution derived from Earth Observation satellites. Specifically, the high-resolution, high-frequency multispectral imagery provided by the Sentinel-2 mission enables the construction of continuous Satellite Image Time Series (SITS). By monitoring how specific agricultural parcels reflect light over time, deep learning models can track the unique phenological growth stages of varying vegetation to classify crop types. Despite these advancements, employing sequence modeling for SITS on a national or global scale presents severe technical and agronomic challenges. Satellite imagery is inherently irregular due to persistent cloud cover, atmospheric interference, and varying sensor revisit times, resulting in noisy sequences with unequal temporal gaps. Furthermore, processing high-dimensional spatial-temporal grids creates critical memory and input/output (I/O) bottlenecks in High-Performance Computing (HPC) environments, severely limiting the scalability of operational crop mapping frameworks.

Beyond hardware constraints, a fundamental limitation of many current state-of-the-art sequence models is their "temporal myopia". These architectures typically treat each growing season as an entirely isolated event, attempting to classify crops based on just one year of weather and growth patterns. In doing so, they completely ignore a foundational rule of agronomic management: crop rotation. Farmers intentionally alternate specific crop species across consecutive years to maintain soil health, prevent nutrient depletion, and break pest cycles. If a satellite observation is heavily obscured by clouds or spectrally ambiguous in a given year, a single-year model lacking inter-year context has no historical background to help it classify the vegetation. Conversely, a model capable of tracking multi-year rotational dependencies can analyze what the farmer planted in the previous and subsequent years to probabilistically infer the missing crop.

This thesis evaluates a machine learning framework designed to navigate hardware limitations while simultaneously capturing both intra-year phenology and inter-year crop rotation dynamics. To overcome the severe HPC bottlenecks associated with large-scale spatial processing, the framework transitions the data pipeline to a highly optimized 1D parcel-aggregated extraction method. By utilizing a 10-meter inward erosion buffer to eliminate mixed boundary pixels, and reducing the data representation to the deep temporal evolution of the spatial mean of individual agricultural fields, the architecture completely bypasses network I/O limits. This dimensional reduction enables the rapid processing of a massive six-year continuous SITS dataset derived from the Austrian INVEKOS (Integrated Administration and Control System) registry, encompassing approximately 300,000 crop fields across 47 distinct agricultural classes. To support rigorous architectural evaluation, the project utilizes an automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna to facilitate fair, distributed hyperparameter tuning.

The research systematically establishes robust single-year baselines to serve as standard benchmarks for phenological extraction. These include a 1-Dimensional Convolutional Neural Network (1D-CNN) and standalone Transformer Encoders evaluated in both generative and discriminative variants. Additionally, to handle irregular temporal gaps specifically within the Transformer models, the framework implements an established AgroDays Positional Encoding approach. Drawing on state-of-the-art techniques in remote sensing, this method adapts standard sinusoidal encodings using the exact agricultural day of the satellite acquisition to explicitly inject temporal distance into the spectral embeddings. Crucially, the research also introduces linear structural controls; in these tests, the deep non-linear feature extractors are replaced by a zero-memory linear backbone that temporally mean-pools the sequence and feeds directly into multi-year sequence heads. This deliberate structural

control motivates investigating the necessity of utilizing high-capacity, non-linear neural backbones before complex multi-year sequence smoothing can be effectively applied to raw satellite data.

To bridge the gap between intra-year phenological tracking and historical agricultural dependencies, two distinct multi-year hybrid architectures are compared. The established Transformer-HMM cascade acts as the Bayesian reference model. In this framework, the Transformer acts exclusively as the emission model, generating yearly class likelihoods, while a Hidden Markov Model (HMM) models the temporal sequence using explicit probabilistic transition matrices empirically initialized from the dataset. Compared with this, the thesis proposes a novel Hierarchical Transformer-LSTM, a purely neural end-to-end architecture. This proposed model processes intra-year sequences in parallel using self-attention, extracting semantic yearly predictions, and feeds this chronological sequence into an inter-year Bidirectional LSTM. By confronting explicit probabilistic transition modeling against deep recurrent memory, this thesis evaluates whether a network can intrinsically learn the latent, non-linear representations of multi-year crop rotations without relying on explicitly defined Markovian state transitions.

Experimental results across the 47 distinct crop classes demonstrate the severe limitations of single-year models when confronted with highly imbalanced agricultural datasets. The 1D-CNN and standalone Discriminative Transformer achieved limited Macro F1-scores of 30.17% and 33.18%, respectively, frequently defaulting to majority staple crops at the expense of minority classes. The linear structural controls proved the necessity of non-linear feature extraction, suffering significant performance degradation with the Linear-LSTM configuration yielding a Macro F1-score of merely 1.04%. Ultimately, both multi-year architectures, evaluated across the full six-year sequence, substantially outperformed the single-year baselines. This demonstrates that historical crop rotation context is highly effective at rescuing minority classes that traditional models ignore. The established Generative Transformer-HMM achieved an optimal Macro F1-score of 63.95%. The novel Generative Transformer-LSTM successfully learned multi-year agricultural cycles end-to-end, reaching a Macro F1-score of 40.12%, marking a clear improvement over its single-year counterpart.

While the LSTM suggests that recurrent memory cells can intrinsically capture temporal dependencies to improve accuracy, it ultimately fell short of the HMM's performance. This performance gap suggests that, although unconstrained recurrent networks may learn multi-year transition rules end-to-end, they could remain highly susceptible to the dataset's extreme class imbalance. Conversely, the HMM's explicit empirical initialization of transition matrices may act as a robust probabilistic prior. This statistical foundation could help protect minority crop transitions from being overshadowed during training, potentially delivering more effective temporal smoothing across complex agricultural cycles. However, further interpretability studies would be required to confirm this mechanism. By evaluating this optimized software framework, this thesis provides a computationally feasible solution for operational longitudinal crop mapping. The findings lay the groundwork for future research, which aims to improve scalability and context integration through the exploration of linear-time State Space Models (SSMs) like Mamba, the reintroduction of spatial structures via Graph Neural Networks (GNNs), and the fusion of multimodal, weather-independent Synthetic Aperture Radar (SAR) data.

1 Introduction and Background

1.1 Introduction

The global agricultural sector is currently facing unprecedented challenges. Driven by rapid population growth, climate change, and the need for sustainable resource management, modern agriculture must maximize crop yields while minimizing environmental impact [1]. Precision agriculture [2, 3] has emerged as a crucial solution to address these demands, relying heavily on accurate, large-scale, and up-to-date information regarding land use and crop distribution. Traditionally, acquiring this data required costly and time-consuming manual field surveys, which are impossible to scale effectively at a national or global level.

Over the past decade, Earth Observation (EO) satellites have revolutionized how we monitor agricultural lands. Constellations such as the European Space Agency's Sentinel-2 provide free, high-resolution multispectral imagery with frequent revisit times (every 5 to 10 days) [4]. By observing how a specific parcel of land reflects light over time [5], it is possible to track the phenological cycle (the biological growth stages) of vegetation. This continuous stream of data allows researchers to identify specific crop types based on their unique growth signatures rather than relying on a single, static snapshot [6].

Simultaneously, the rise of Deep Learning (DL) has provided the computational tools necessary to process this massive influx of satellite data. Early approaches in remote sensing relied heavily on traditional machine learning algorithms and spatial pattern recognition. However, as the field evolved, it became evident that the temporal dimension (how a pixel changes over a year) holds the most predictive power for crop classification. Consequently, sequence-modeling architectures such as Temporal Convolutional Neural Networks (1D-CNNs) [7, 8] and Recurrent Neural Networks (RNNs) [9] have become the standard for agricultural time-series analysis.

Despite these advancements, operationalizing deep learning for crop mapping presents significant difficulties. First, Satellite Image Time Series (SITS) are inherently irregular; optical sensors cannot penetrate clouds, leading to missing data and uneven temporal gaps that disrupt traditional machine learning models. Second, processing high-dimensional spatial-temporal datasets (such as tracking entire 2D image patches over a temporal sequence) creates severe computational and memory bottlenecks, particularly in High-Performance Computing (HPC) environments. Finally, most current state-of-the-art models suffer from *"temporal myopia"*; they analyze only a single year of weather and phenology, completely ignoring the fundamental agricultural practice of crop rotation, where farmers intentionally alternate crops across multiple years to maintain soil health [10].

This thesis aims to bridge these gaps. To overcome the hardware bottlenecks associated with heavy spatial-temporal grids, this work shifts to a highly optimized 1D pixel-based pipeline. By analyzing the temporal evolution of the spatial mean of entire agricultural fields rather than broad 2D spatial patches, the computational load is drastically reduced, enabling rapid prototyping and extensive hyperparameter optimization on an HPC cluster. Building upon this efficient infrastructure, the project establishes a comprehensive suite of robust baselines. Single-year intra-year models (specifically 1D-CNNs and standalone Transformer Encoders) aretuned to serve as standard phenological benchmarks. Alongside these, simplified baseline models pairing a basic linear layer directly with standalone LSTM [11] and HMM [12] heads are evaluated. These configurations serve as a deliberate benchmark test, demonstrating the role of high-capacity, non-linear feature extractors for sequence models, in particular the multi-year probabilistic smoothing performed by the HMM, to be able to effectively interpret raw satellite data.

These baselines are systematically compared against two high-capacity, multi-year hybrid architectures. The first is the Transformer-HMM, an established state-of-the-art Bayesian framework [13]

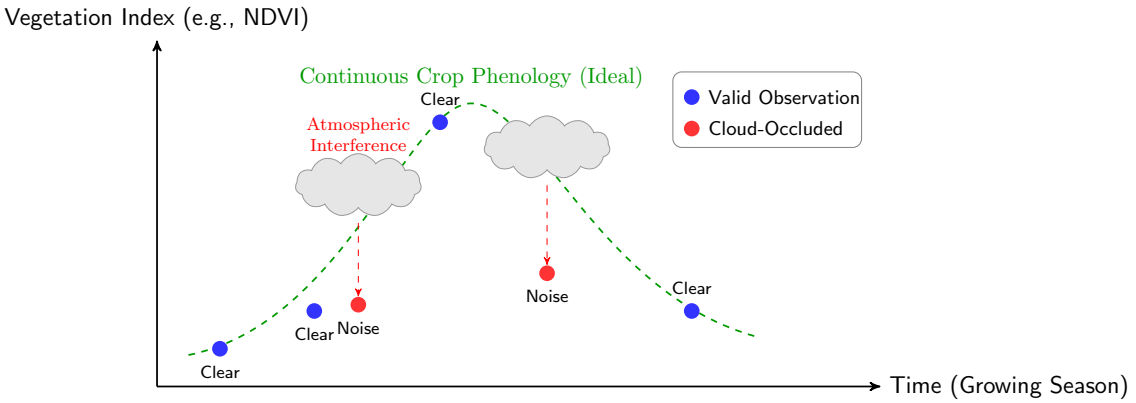


Figure 1.1: Conceptual illustration of a Satellite Image Time Series (SITS) capturing crop phenology. The dashed line represents the continuous, ideal growth cycle of the crop on the ground. In contrast, the points represent the discrete Sentinel-2 acquisitions. Cloud cover frequently occludes the optical sensor, creating irregular temporal gaps and noisy measurements that sequence models must navigate.

utilized here as an advanced reference model. To optimize the emission probabilities provided to the HMM transition layer, different backbone architectures are explicitly compared—including 1D-CNNs and both generative and discriminative variants of the Transformer backbone [13]. Although state-space models like Mamba [14] were originally slated for this feature-extraction comparison, time constraints limited the executed backbones to convolutions and self-attention. The second is a novel, purely neural end-to-end architecture proposed by this thesis: the Hierarchical Transformer-LSTM. By integrating the self-attention mechanism proven effective on satellite time series [15, 16] with the recurrent memory of an LSTM, this research systematically confronts Bayesian transition modeling against deep recurrent memory to answer whether neural networks can intrinsically learn and leverage the historical rules of multi-year crop rotation to significantly outperform models that rely on single-year phenology alone.

1.2 Research Question and Objectives

The goal of this thesis is to explore the efficacy of deep sequence-modeling architectures in classifying agricultural crop types using irregular SITS. Specifically, this research investigates the transition from traditional, single-year phenological analysis to a multi-year, hybrid approach capable of learning long-term farming practices. This leads to the primary research question of this study:

> *"How can we efficiently process irregular Satellite Image Time Series for large-scale crop mapping, and does the integration of multi-year crop rotation history via hybrid Transformer-LSTM and Transformer-HMM architectures significantly outperform traditional single-year phenological models?"*

To systematically answer this question, the project is structured around four specific objectives, spanning both machine learning infrastructure and architectural design:

i) **Overcoming I/O and Computational Bottlenecks:** To redesign the data pipeline, shifting from heavy spatial-temporal patches to a highly optimized 1D pixel-based extraction method. This aims to resolve severe High-Performance Computing (HPC) memory constraints and reduce epoch training times, enabling rapid experimentation.

ii) **Establishing Robust Control Baselines:** To design, train, and tune standard non-hybrid temporal architectures to serve as phenological benchmarks. Specifically, a 1-Dimensional Convolutional Neural Network (1D-CNN) and standalone Transformer Encoders—evaluated in both generative and discriminative variants—are explicitly compared to determine optimal backbone feature extraction capabilities. While state-space models such as Mamba were initially intended to be part of this backbone comparison, time and computational constraints precluded

their execution. Alongside these, standalone LSTM and HMM configurations utilizing a zero-memory linear backbone are executed. These provide a deliberate structural control test, evaluating the role of high-capacity non-linear feature extractors for raw temporal data before multi-year sequence smoothing can be applied.

iii) **Developing Multi-Year Hybrid Architectures:** To construct the novel Hierarchical Transformer-LSTM and confront it against the established Transformer-HMM to evaluate sequence modeling beyond a single harvest cycle. This objective focuses on using self-attention to manage intra-year weather irregularities, while comparing an LSTM memory cell against an HMM cascade to capture inter-year crop rotation dependencies. Additionally, both generative and discriminative variants of the Transformer backbone are tested to optimize the emission probabilities passed to the sequence heads.

iv) **Automating Hyperparameter Optimization:** To build a robust, fully automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna. This ensures all models are evaluated fairly at their optimal identified configurations through distributed hyperparameter tuning on an HPC cluster.

1.3 State of the Art

1.3.1 Deep Learning for Precision Agriculture

The paradigm of precision agriculture has fundamentally transformed modern farming management. By transitioning from uniform field-level treatments to highly granular, site-specific interventions, precision agriculture seeks to optimize resource inputs—such as water, fertilizers, and pesticides—while maximizing overall crop yield [2, 3]. This shift relies entirely on the continuous acquisition and analysis of vast amounts of heterogeneous data collected from on-site Internet of Things (IoT) sensors, Unmanned Aerial Vehicles (UAVs), and Earth Observation (EO) satellites.

Historically, extracting actionable insights from these diverse data sources depended on traditional machine learning algorithms, such as Random Forests (RF) and Support Vector Machines (SVM) [17, 18]. While effective for specific, localized tasks, these traditional methods often rely on manual feature engineering. Researchers were required to construct and select specific vegetation indices or spatial textures beforehand, a process that is highly domain-specific, time-consuming, and difficult to scale across different geographical regions and climatic conditions.

The advent of Deep Learning (DL) has largely overcome these limitations, triggering a major shift in agricultural data analysis [19]. Unlike traditional machine learning, deep neural networks possess the capacity for automatic, hierarchical feature extraction. By ingesting raw or minimally processed data, DL models can autonomously learn complex, non-linear representations directly from the input space. In precision agriculture, this has led to unprecedented breakthroughs across a wide array of applications, including automated weed detection, early disease diagnosis, and precise yield estimation [20].

In the specific context of remote sensing and land cover mapping, deep learning has proven exceptionally proficient at handling the massive dimensionality of Satellite Image Time Series (SITS). By leveraging specialized architectures capable of modeling both spatial hierarchies and temporal dependencies, deep learning models can now accurately map crop distributions at a national scale, frequently achieving higher accuracies than traditional shallow classifiers and forming the foundational technology upon which this thesis is built [7].

1.3.2 Remote Sensing and Sentinel-2 Time Series

The European Space Agency's Sentinel-2 mission has revolutionized agricultural monitoring by providing high spatial resolution imagery with a frequent 5-day revisit time. This creates dense Satellite Image Time Series (SITS) that capture the vital phenological evolution of crops throughout the growing season [21]. However, extracting actionable insights from SITS requires models to navigate a complex time-space tradeoff. Research demonstrates that for crop type classification on Sentinel-2 data, the temporal structure is often richer than the spatial structure; consequently, hybrid models (like recurrent

convolutional networks) perform best when the majority of their parameters are allocated to modeling temporal sequences rather than spatial features [21].

To effectively process these time series, researchers have explored various representation and sequence-modeling strategies:

- **Image-Based Encodings:** One approach involves transforming 1D time-series data (such as vegetation indices) into 2D image representations (like Gramian Angular Fields or Markov Transition Fields) allowing standard convolutional networks to efficiently extract static and dynamic temporal patterns [22].
- **Recurrent to Transformer Evolution:** While models like Long Short-Term Memory (LSTM) networks have traditionally handled sequential data, modern approaches increasingly rely on Transformer architectures to overcome LSTM limitations with long-range dependencies [23]. In crop mapping, advanced models like the Time-Series Vision Transformer (TSVIT) utilize purely self-attention mechanisms to process multi-modal SITS, combining optical data (Sentinel-2) with radar (Sentinel-1) to boost classification performance without relying on standard convolutions [24].

Despite the rich information in SITS, operational mapping is frequently hindered by cloud cover noise and a severe lack of pixel-level ground truth data. Recent deep learning innovations address these bottlenecks directly:

- **Knowledge-Guided Networks:** To ensure models learn meaningful phenology rather than dataset artifacts, prior domain knowledge can be integrated into the architecture. For instance, the Knowledge-Guided Crop Mapping Network (KGCMNet) forces the model to reconstruct known vegetation indices (like NDVI) during training, which significantly sharpens its spatio-temporal representations [25].
- **Weakly Supervised Learning:** To bypass the need for exhaustive pixel-level annotations, weakly supervised frameworks (e.g., Exact) utilize only image-level labels. These models explore space-time perceptive clues to rectify erroneous semantic biases caused by anomalous temporal clips, pushing weakly supervised performance to nearly 95% of fully supervised baselines [26].
- **Multimodal Pseudo-Labeling:** Another highly scalable solution involves integrating top-down SITS with bottom-up, ground-level field images. By using multimodal data to generate reliable pseudo-labels, models can achieve high accuracy across vast geographical areas while drastically reducing the need for expensive on-site data collection [27].

1.3.3 Sequence Modeling: From CNNs to Hybrid Architectures

The accurate classification of agricultural land relies heavily on the ability to interpret the chronological sequence of crop phenology. As deep learning has evolved, so too have the architectures designed to model these complex temporal dynamics.

Early deep learning approaches for Satellite Image Time Series (SITS) relied heavily on Convolutional Neural Networks (CNNs). While originally designed to extract spatial features from static images [20], CNNs were successfully adapted for temporal sequences via 1-Dimensional (1D) convolutions. By operating filters across the temporal dimension of a pixel's profile, 1D-CNNs can efficiently capture localized temporal patterns and phenological shifts, frequently outperforming traditional machine learning classifiers like Random Forests [7] in large-scale temporal benchmarks. However, standard CNNs lack an inherent mechanism to model strict, long-term sequential dependencies, as their receptive field is limited by the kernel size, unless compensated by depth, dilation, or specific architectural adaptations.

To explicitly model the chronological order of agricultural events, Recurrent Neural Networks (RNNs) became the standard. The foundational RNN Encoder-Decoder architecture allowed models to ingest variable-length sequences and maintain a hidden state that summarizes past inputs to inform future predictions [28]. To address the vanishing gradient problem inherent in basic RNNs, Long Short-Term Memory (LSTM) networks were widely adopted. These architectures are particularly

advantageous in remote sensing, as specialized recurrent layers can be engineered to handle the irregular time gaps, cloud occlusions, and missing values that commonly afflict satellite time series [29].

Despite their success, LSTMs process data sequentially, which bottlenecks training parallelization and limits their ability to capture extremely long-range dependencies across a full growing season [23]. The introduction of Transformer architectures marked a paradigm shift in sequence modeling. By dispensing with recurrence and relying entirely on self-attention mechanisms, Transformers evaluate the relationships between all time steps simultaneously, regardless of their distance in the sequence [23]. In modern crop mapping applications, Transformers have demonstrated superior classification accuracy and generalizability compared to 1D-CNNs, especially when handling complex, multi-year, or irregularly sampled temporal features [30].

Crucially, the output formulation of the Transformer plays a significant role when interfacing with probabilistic sequence models like HMMs. Transformers can be configured discriminatively (using a standard linear layer with softmax) to estimate class posterior probabilities directly, or generatively (using a normalized softmax layer with no bias term) to implicitly enforce a uniform class prior [13]. The generative approach reduces bias toward dominant classes and aligns perfectly with the emission model requirements of Bayesian transition networks [13].

To maximize predictive performance, researchers have increasingly developed hybrid architectures that integrate multiple sequence-modeling paradigms. Early hybrid networks in remote sensing frequently combined CNNs with RNNs (such as ConvLSTMs or CNN+GRU architectures) to simultaneously extract spatial textures and temporal phenology [21]. As the focus has shifted toward capturing highly complex, multi-scale temporal dynamics, newer state-of-the-art approaches merge the localized sequential extraction capabilities of LSTMs with the global, long-range processing power of Transformers [31, 23].

While these standard hybrid models represent the current benchmark for decoding spatiotemporal Earth observation data, they typically process sequences within a single temporal scale. This thesis proposes a novel inversion of this paradigm specifically tailored for multi-year agricultural monitoring. Rather than using both networks on the same sequence, the proposed architecture separates the timescales: Transformers are utilized to concurrently extract dense, intra-year phenological features, while Bidirectional LSTMs (as proposed by this thesis) or Hidden Markov Models (as established by recent literature) are subsequently deployed to capture the long-term, inter-year sequential dependencies of crop rotations.

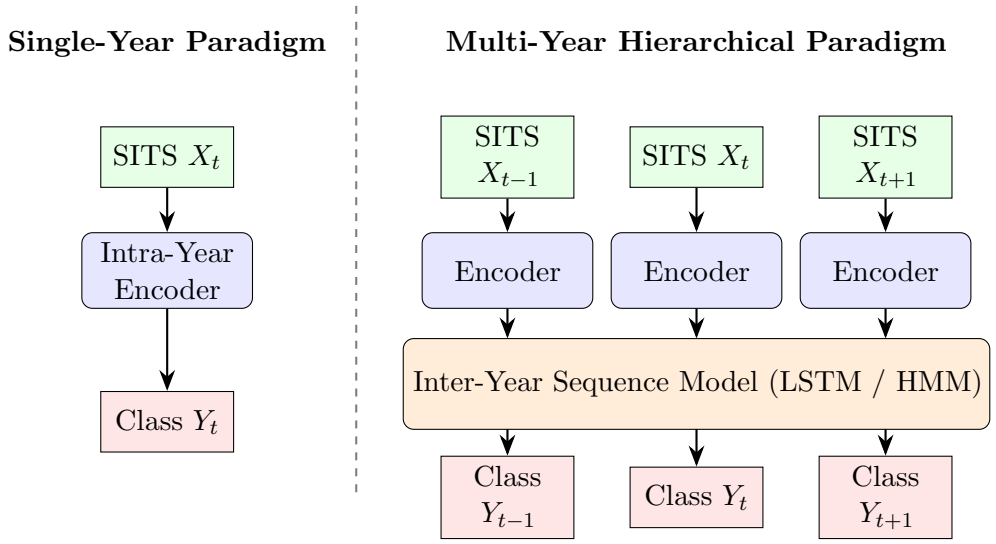


Figure 1.2: Conceptual comparison between single-year sequence processing and hierarchical multi-year architectures. The shared horizontal layer illustrates the integration of inter-year transition modeling.

1.3.4 Multi-Year Crop Rotation Modeling

The Importance of Long-Term Agricultural Dynamics

Crop rotation is a fundamental agricultural practice wherein farmers alternate the species of crops

grown on a specific plot across consecutive seasons. This strategy is essential for maintaining soil health, preventing nutrient depletion, and breaking pest and disease cycles. Despite its agronomic importance, most remote sensing models suffer from "temporal myopia," i.e., analyzing single-year phenology while ignoring inter-annual historical dependencies.

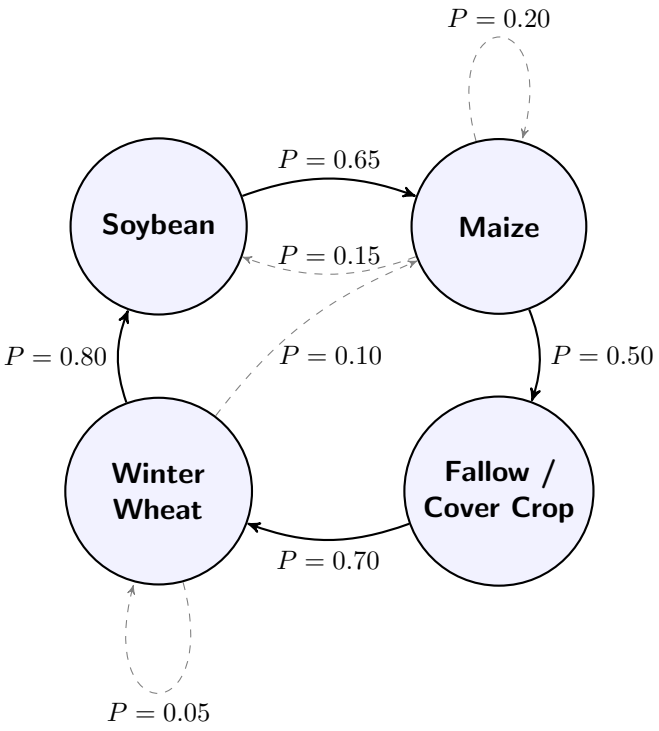


Figure 1.3: Crop rotation represented as a probabilistic state transition graph. Multi-year sequence models aim to intrinsically learn these transitional dependencies (edges) to improve classification accuracy when emission likelihoods (nodes) are uncertain.

Traditional and Knowledge-Based Approaches

Early attempts to map multi-year dynamics relied on explicitly injecting prior agronomic knowledge into the classification pipeline. For example, probabilistic models such as Bayesian networks and Conditional Random Fields have been used to encode known crop rotation rules; the outputs of these networks are then fed into standard classifiers (such as Support Vector Machines) to guide the algorithm and improve early-season mapping accuracy [17]. While effective, these methods require manually constructed rule bases and lack the flexibility to adapt to unexpected crop sequence changes.

Machine Learning and Temporal Transferability

As satellite archives have grown, researchers have increasingly utilized machine learning to track land-use transitions over longer periods. Hierarchical classification workflows (such as tiered Random Forests) have been successfully deployed across multiple years to explicitly track crop rotations, field abandonment, and land-use intensity [18]. However, monitoring annual crop rotations over a decade presents significant challenges regarding temporal transferability; models trained on a specific year often struggle to generalize to past or future years due to shifting weather patterns, varying sensor availability, and inter-annual phenological shifts [30].

The Deep Learning Frontier

While early deep learning methods successfully tracked multi-year changes, they predominantly processed each year independently and relied on post-classification comparisons to identify rotations. The next frontier in precision agriculture mapping is the development of architectures that possess the intrinsic memory capacity to learn and predict multi-year agricultural cycles directly from continuous, raw satellite time-series data.

Recently, researchers have demonstrated significant success in bridging these paradigms by integrating Bayesian modeling—such as Hidden Markov Models (HMMs)—with deep Transformer Encoders to explicitly capture intricate multi-year crop rotation patterns [13]. In this cascade setup, the Transformer

acts as the emission model providing class likelihoods, while the HMM acts as the transition model. The HMM models the temporal sequence through explicit transition matrices. Rather than relying strictly on fixed agronomic rules, these matrices can be initialized empirically from dataset frequencies and jointly optimized during training, allowing the model to balance statistical temporal coherence with the emission likelihoods provided by the Transformer. Building upon these foundations, the focus now shifts toward confronting these probabilistic cascades with purely neural hybrid architectures, such as LSTMs, capable of modeling complex temporal dependencies end-to-end without relying on explicitly defined Markovian state transitions.

1.4 Contributions

This thesis advances the field of agricultural remote sensing by addressing the computational bottlenecks of large-scale mapping and exploring the predictive power of multi-year sequence modeling. The primary contributions of this work are as follows:

- **Adoption of an Optimized 1D Data Pipeline:** To navigate the limitations of memory-intensive 2D spatial patches observed during initial exploratory phases, a streamlined, 1D parcel-aggregated data representation is utilized. This approach bypasses severe input/output (I/O) bottlenecks, rendering the training of multi-year models computationally feasible on a single workstation.
- **Comprehensive Baseline Benchmarking:** Robust single-year temporal baseline models—specifically a **1-Dimensional Convolutional Neural Network (1D-CNN)** and standalone **Transformer Encoders** (evaluated in both generative and discriminative variants)—are developed and tuned to provide a standard for intra-year phenological classification. Additionally, standalone tests using a simplified **linear backbone** paired with **LSTM** and **HMM** heads are executed. These serve as a deliberate control mechanism, proving the structural necessity of high-capacity non-linear feature extractors when processing raw SITS.
- **Confronting Novel and Established Multi-Year Architectures:** A custom Hierarchical **Transformer-LSTM** architecture is designed and compared to the established **Transformer-HMM** Bayesian approach. This comparison evaluates whether deep recurrent memory or probabilistic transition matrices better capture complex inter-year crop rotation dependencies. Furthermore, both generative and discriminative variants of the Transformer are evaluated to determine the optimal configuration for generating emission probabilities.
- **Integration of Automated MLOps and Distributed Tuning:** A robust Machine Learning Operations (MLOps) workflow, leveraging PyTorch Lightning, Ray Tune, and Optuna, is adapted to automate the training pipeline. This infrastructure facilitates distributed hyperparameter optimization across an HPC cluster, ensuring that all architectures are evaluated fairly and at their maximum potential.

By utilizing this streamlined framework, this thesis provides a computationally feasible software pipeline for SITS analysis, offering critical insights into whether deep sequence models can intrinsically learn the historical, multi-year rules of agricultural crop rotation.

2 Methodology and Architectural Design

2.1 Single-Year and Structural Baselines

To establish a rigorous benchmark for multi-year architectures, a comprehensive suite of single-year sequence models and structural controls was developed. These networks are restricted to processing intra-year phenological data, providing a comparative standard to justify the complexity of multi-year modeling.

2.1.1 Temporal 1D Convolutional Neural Network (1D-CNN)

The first phenological baseline is a 1-Dimensional Convolutional Neural Network (1D-CNN). By sliding convolutional filters across the time dimension of a pixel's spectral profile, the network efficiently extracts localized temporal features and phenological shifts, a methodology that has proven highly effective for satellite time series classification [8].

The encoder consists of two sequential convolutional blocks. The initial layer maps the 13 input satellite bands to 64 channels using a kernel size of 3 and a padding of 1 to preserve the temporal sequence length. This is followed by a 1D Batch Normalization layer and an in-place ReLU activation to introduce non-linearity. The second layer further projects the features from 64 to 128 channels, utilizing the same kernel size, padding, normalization, and activation parameters.

For classification, a point-wise convolution (kernel size of 1) is applied to map the extracted 128-dimensional features to the target crop classes. Because the network produces predictions for each time step, a **masked** temporal mean pooling of the unnormalized logits is executed across all time steps in the year to generate the final field-level classification, explicitly ensuring that zero-padded tokens are excluded from the average. It is important to note that with two convolutional layers utilizing a kernel size of 3, a stride of 1, and a padding of 1, the local receptive field spans exactly five consecutive observations. Because the input sequence is irregularly sampled due to cloud cover, these five observations do not correspond to a fixed number of days, meaning the CNN relies strictly on sequence order rather than exact temporal distances.

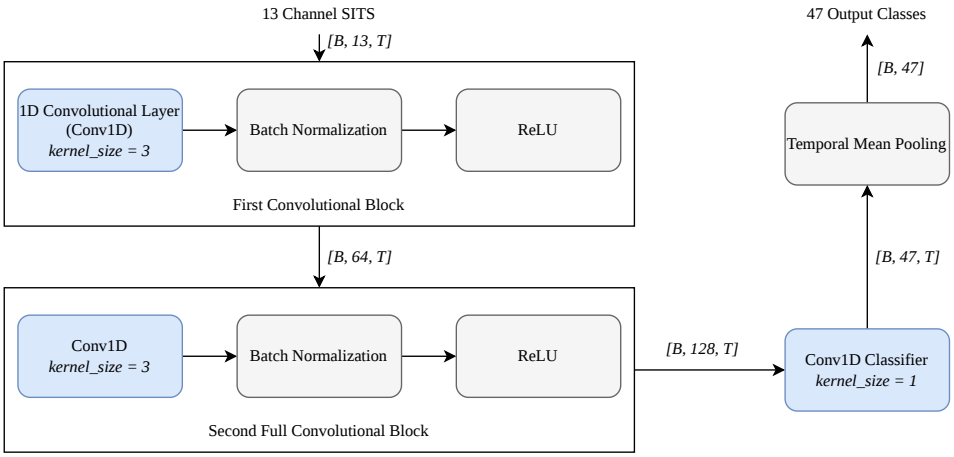


Figure 2.1: Architectural block diagram of the Temporal 1D-CNN baseline.

2.1.2 Standalone Transformer Encoders (Generative and Discriminative)

To capture global temporal dependencies without being restricted to the narrow, sliding time windows of a CNN, a standalone **Transformer Encoder** serves as the second primary baseline. Utilizing Multi-Head Self-Attention [15], this architecture evaluates the entire yearly sequence simultaneously, mapping the complex growth stages of crops into dense representations [32].

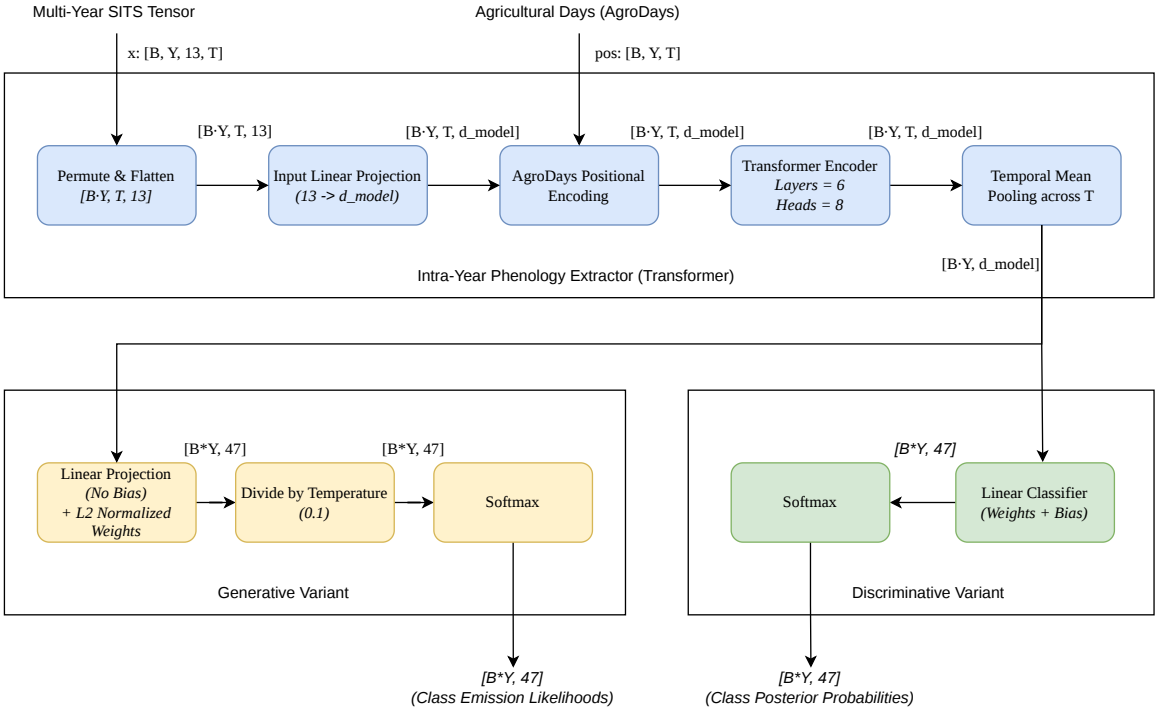


Figure 2.2: Architectural block diagram of the standalone Transformer Encoder (Generative and Discriminative variants).

Crucially, the output formulation of the Transformer is evaluated in two distinct variants to benchmark their efficacy as both standalone classifiers and future emission models:

- **Discriminative Variant:** Employs a standard classification layer to directly predict the posterior probability $P(c|x)$ of each crop class c based on the input sequence representation h. This is formulated using a standard linear projection with bias, followed by a softmax function:

$$P(c|x) = \frac{\exp(w_c^T h + b_c)}{\sum_{k=1}^{K} \exp(w_k^T h + b_k)}$$

  where w_c and b_c are the weights and bias for class c, and K is the total number of classes.

- **Generative Variant:** Utilizes a specifically modified, normalized output layer. This design treats all crop types as equally likely by default, which prevents the network from unfairly favoring the most common crops in the dataset [13]. It achieves this by removing the bias term b_c and normalizing the weights, often incorporating a temperature scalar τ to sharpen the emission likelihoods:

$$P(c|x) \propto \frac{\exp(\frac{1}{\tau} w_c^T h)}{\sum_{k=1}^{K} \exp(\frac{1}{\tau} w_k^T h)}$$

  More importantly, generating these probabilities without prior bias matches the exact mathematical format required by the HMM to effectively map sequence transitions [13].

2.1.3 AgroDays Positional Encoding

Agricultural satellite time series are inherently irregularly sampled due to cloud cover and varying satellite revisit times. Standard sequential positional encodings would falsely inform a self-attention

mechanism that all observations are equidistant. To address this, all Transformer variants in this study (both the single-year baselines and the multi-year hybrid architectures) utilize an AgroDays Positional Encoding module.

Instead of indexing observations as sequential integer steps, this module adapts the standard sinusoidal positional encoding [15] by substituting the sequence index with the exact agricultural day of the acquisition. This adaptation is a well-established technique in remote sensing time series analysis [32, 16]. While the Euclidean distance between two sinusoidal encodings is a periodic function and therefore does not guarantee a strictly linear, proportional preservation of temporal distance, this mapping effectively makes the actual acquisition dates explicitly available to the model.

Algorithm 1 AgroDays Positional Encoding (Lookup Table)

Require: Maximum agricultural day $max_seq \leftarrow 366$
Require: Pre-compute during initialization:
$table \leftarrow \text{zeros}(max_seq, d_model)$
for $pos = 0$ **to** $max_seq - 1$ **do**
  for $i = 0$ **to** $(d_model/2) - 1$ **do**
    $freq \leftarrow pos/10000^{2i/d_model}$
    $table[pos, 2i] \leftarrow \sin(freq)$
    $table[pos, 2i+1] \leftarrow \cos(freq)$
  end for
end for
$PE_Layer \leftarrow \text{FrozenEmbedding}(table)$
Require: Forward Pass:
Require: Input sequence $X \in \mathbb{R}^{B \times T \times d_model}$
Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times T}$
Ensure: Position-encoded sequence $X_{enc} \in \mathbb{R}^{B \times T \times d_model}$
$PE \leftarrow PE_Layer(pos)$ ▷ Look up pre-computed vectors using day index
$X_{enc} \leftarrow X + PE$ ▷ Inject temporal signal into embeddings
return X_{enc}

As detailed in Algorithm 1, the position-aware embeddings are pre-computed during the network's initialization. A geometric progression of wavelengths is calculated across the network's embedding dimension and stored within a frozen lookup table [15]. During the forward pass, the model simply uses the input matrix *pos* as indices to extract the correct sine and cosine vectors for each specific agricultural day, injecting this explicit temporal signal directly into the spectral embeddings.

To handle calendar variations such as leap years, the day index is bounded to a 366-day cycle. For days lacking valid satellite acquisitions due to irregular revisit times, no explicit temporal padding is required for the positional encoding itself; instead, the overall sequence is zero-padded to a maximum length, and a standard padding mask is supplied to the Transformer to ensure the attention mechanism ignores these artificial tokens.

2.1.4 Linear Structural Controls

To scientifically validate the necessity of the aforementioned non-linear backbones, standalone multi-year controls utilizing a zero-memory linear baseline were evaluated. Rather than relying on randomly initialized weights, this control relies on a fully trainable linear layer devoid of intermediate non-linearities.

Specifically, the raw 13-band SITS data is temporally mean-pooled across the growing season to completely collapse the intra-year temporal sequence into a static feature vector. A single trained linear layer then maps these 13 features directly to the 47 output classes. This projection is jointly optimized alongside the sequence heads (LSTM and HMM) utilizing a standard Categorical Cross-Entropy loss. Because a single linear layer applied to temporally pooled data cannot mathematically disentangle highly overlapping, non-linear phenological profiles, it outputs near-random temporal features. Furthermore, because this control utilizes a temporal mean, it entirely collapses the intra-annual temporal sequence. Therefore, this configuration evaluates the dual impact of removing both non-linear feature extraction

and the chronological intra-year temporal structure. Testing the sequence heads on these impoverished inputs isolates their transition-modeling capabilities, explicitly evaluating whether multi-year sequence smoothing requires a high-capacity, non-linear feature extractor to function.

2.2 Multi-Year Architectures

The core focus of this thesis is the development and evaluation of hybrid, multi-scale neural architectures capable of simultaneously modeling intra-year phenological dynamics and inter-year crop rotation patterns.

2.2.1 The Reference Transformer-HMM

To evaluate purely neural multi-year models, an established state-of-the-art framework is required for comparison. This research implements the Bayesian cascade architecture proposed by Perantoni et al. [13], which integrates a Transformer Encoder with a Hidden Markov Model (HMM).

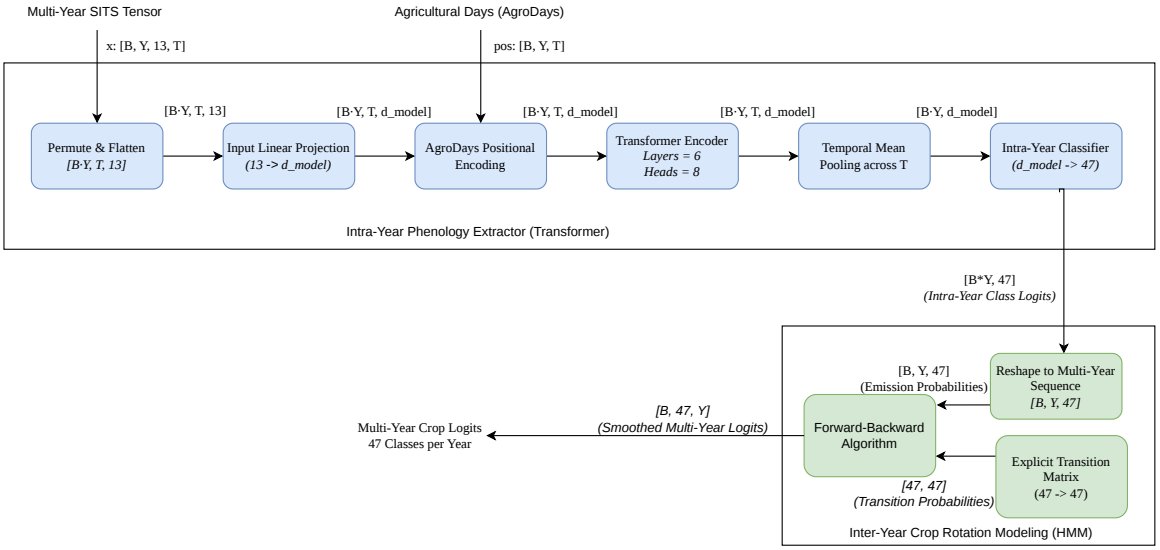


Figure 2.3: Hierarchical overview of the Transformer-HMM approach, demonstrating the fusion of intra-year attention and inter-year transition modeling.

In this hierarchical setup, the Transformer acts exclusively as the emission model, processing the intra-year SITS to estimate the emission likelihoods of each crop class for a given year. These likelihoods are then passed into the HMM transition layer. To ensure a scientifically rigorous comparison against the purely neural recurrent baseline, it is crucial to detail the exact formulation of the HMM implemented in this thesis [13].

Rather than relying on frozen, manually defined agronomic rules, the HMM's explicit probabilistic transition matrices are jointly optimized during training alongside the neural backbone [13]. The transition model is formulated as a first-order, time-variant Markov chain; this means a distinct transition matrix is utilized for each specific yearly interval (e.g., $T_{2016\to2017}$ is independent of $T_{2017\to2018}$) to account for inter-annual climate or policy shifts [13]. To ensure realistic starting conditions, the matrices and initial prior distributions are empirically initialized using the exact crop co-occurrence frequencies computed from the training dataset [13]. To handle completely unobserved rotations and guarantee numerical stability, a smoothing constant (1×10^{-20}) is added to all transition counts before mapping them into log-space. This prevents zero-probability transitions from causing catastrophic $-\infty$ log-likelihoods during training. All subsequent computations are executed in log-space using the numerically stable log-sum-exp trick [13].

During training and inference, the architecture employs a forward-backward algorithm to perform marginal smoothing; it computes the posterior probability of each crop class at year t given the entire multi-year sequence of observations $x_{1:T}$. The network's final output logits are derived by combining (summing in log-space) the forward and backward probabilities to compute the smoothed marginals [13].

The network's final output logits are derived by combining the forward and backward probabilities, which mathematically translates to summing their log-probabilities (the log-space equivalent of probability multiplication), to compute the smoothed marginals [13].

While Viterbi decoding (which seeks the single maximum probability sequence $\arg\max_{s_{1:T}} P(s_{1:T}|x_{1:T})$) is also computed as a supplementary metric, the primary sequence optimization relies on the smoothed marginals. Consequently, while the HMM heavily penalizes unlikely crop rotations, it does not enforce rigid, deterministic agronomic constraints; rather, it balances the Transformer's emission likelihoods with maximum statistical temporal coherence.

2.2.2 The Novel Hierarchical Transformer-LSTM

As an alternative to Bayesian transition smoothing, this thesis proposes a novel, purely neural multi-year architecture: the Hierarchical Transformer-LSTM. This model is designed to learn both intra-year phenology and inter-year crop rotations end-to-end, without relying on fixed, explicit transition matrices.

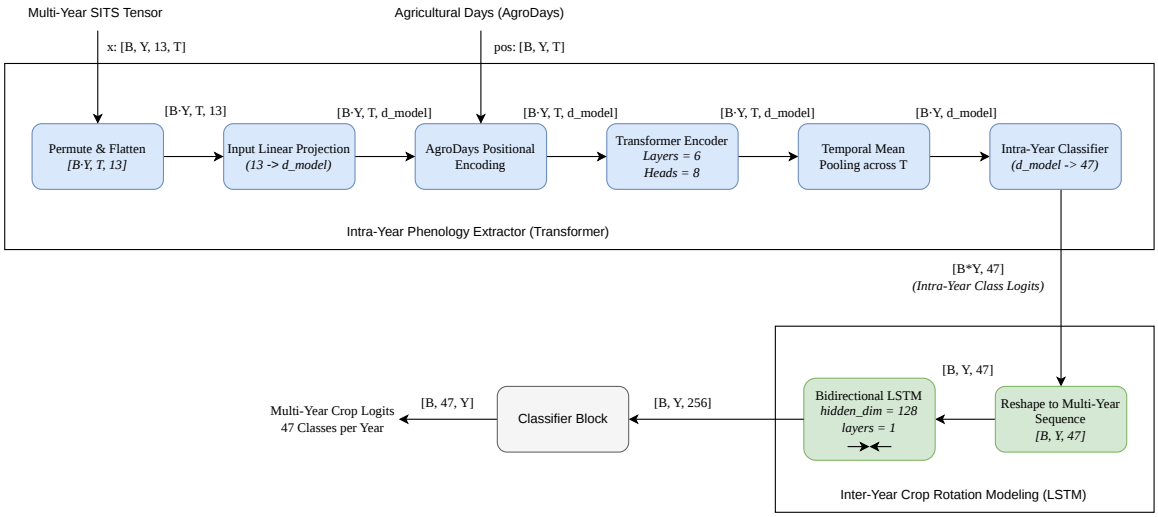


Figure 2.4: Hierarchical overview of the Transformer-LSTM architecture, demonstrating the fusion of intra-year attention and inter-year recurrence.

The architecture processes the temporal scales using a two-stage approach. In the first stage, the sequence is flattened to merge the batch and year dimensions, allowing the Intra-Year Transformer Encoder to evaluate all years independently and concurrently by flattening the temporal sequences into the batch dimension. The Transformer's complex phenological outputs are temporally masked-mean-pooled (strictly excluding artificial padding tokens) and passed through an intra-year classifier, collapsing the dense *d_model* feature extraction into a sequence of real-valued scores (logits) for the 47 specific target classes. Crucially, this intermediate 47-dimensional vector is not merely a latent representation; the Intra-Year Transformer is explicitly optimized using an auxiliary intra-year classification loss during training. This forces the intermediate outputs to represent calibrated semantic class predictions. Furthermore, compressing the high-dimensional embeddings down to 47 semantic logits before recurrence drastically reduces the LSTM's parameter footprint, forcing it to act strictly as an inter-year transition model rather than a redundant feature extractor. While passing the full *d_model* embedding to the LSTM represents a valuable future ablation study, this bottleneck isolates the evaluation strictly to class-level transition dynamics.

In the second stage, this chronological sequence of yearly predictions is restored and passed into an **Inter-Year Bidirectional LSTM** [11].

The bidirectional nature of the LSTM is highly critical; it enables the network to analyze the rotation history in both the forward and backward temporal directions. This allows the model to refine its current crop prediction based on deep inter-year transition rules learned from preceding and subsequent planting decisions. Finally, the concatenated bidirectional hidden states are passed through a dropout regularization layer and a final linear classifier to output the highly contextualized

Algorithm 2 Hierarchical Transformer-LSTM Forward Pass

Require: Multi-year input tensor $X \in \mathbb{R}^{B \times Y \times C \times T}$
Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times Y \times T}$
Ensure: Multi-year crop class logits $Y_{pred} \in \mathbb{R}^{B \times n_classes \times Y}$

Stage 1: Intra-Year Processing (Transformer)
$X \leftarrow \text{permute}(X, [0, 1, 3, 2])$ ▷ Target shape: $[B, Y, T, C]$
$X_{flat} \leftarrow \text{reshape}(X, [B \cdot Y, T, C])$ ▷ Flatten for independent yearly processing
$pos_{flat} \leftarrow \text{reshape}(pos, [B \cdot Y, T])$
$X_{emb} \leftarrow \text{LinearProjection}(X_{flat})$
$X_{emb} \leftarrow \text{AgroDaysPositionalEncoding}(X_{emb}, pos_{flat})$
$T_{out} \leftarrow \text{TransformerEncoder}(X_{emb})$ ▷ Self-Attention over time steps
$Y_{emb} \leftarrow \text{masked_mean}(T_{out}, \text{dim} = 1, \text{mask})$ ▷ Masked temporal pooling, ignoring padding: $[B \cdot Y, d_model]$
$Y_{logits} \leftarrow \text{IntraYearClassifier}(Y_{emb})$ ▷ Map to real-valued logits: $[B \cdot Y, n_classes]$
Stage 2: Inter-Year Processing (LSTM)
$Y_{seq} \leftarrow \text{reshape}(Y_{logits}, [B, Y, n_classes])$ ▷ Restore sequence of years
$L_{out}, (h_n, c_n) \leftarrow \text{BidirectionalLSTM}(Y_{seq})$ ▷ Model crop transitions
$L_{drop} \leftarrow \text{Dropout}(L_{out})$ ▷ Apply temporal regularization
Stage 3: Multi-Year Classification
$L_{logits} \leftarrow \text{InterYearClassifier}(L_{drop})$ ▷ Map LSTM hidden states to final classes
$Y_{pred} \leftarrow \text{permute}(L_{logits}, [0, 2, 1])$ ▷ Final shape: $[B, n_classes, Y]$
return Y_{pred}

unnormalized logit scores for all target crop classes across the entire multi-year sequence.

The entire network is optimized end-to-end using a combined loss function: $\mathcal{L}_{total} = \mathcal{L}_{inter} + \lambda \mathcal{L}_{intra}$, where $\mathcal{L}_{inter}$ is the primary cross-entropy loss from the multi-year sequence output, $\mathcal{L}_{intra}$ is the auxiliary intra-year cross-entropy loss, and λ is a weighting coefficient empirically set to 1.0 to balance feature extraction and transition modeling.

3 Experiments, Datasets, and Infrastructure

3.1 Experimental Setup and Datasets

To rigorously evaluate the proposed multi-year architectures against the established single-year baselines, a standardized experimental framework was designed. All models were trained, validated, and tested using identical data splits to ensure a strictly fair comparative analysis.

3.1.1 The INVEKOS Dataset and Sentinel-2

To evaluate the proposed sequence-modeling architectures, experiments were conducted using data derived from the Austrian **INVEKOS** (Integrated Administration and Control System) registry [33]. INVEKOS serves as the European Union's official spatial registry for agricultural parcel identification. It is important to clarify that this dataset provides administrative reference data based on farmers' declarations, which inherently contains a degree of declarative noise and administrative boundary shifts, rather than representing a flawless, pixel-perfect ground truth. To focus the analysis, the dataset extracts data specifically from the Military Grid Reference System (MGRS) tile 33UVP, chosen for its consistent orbital coverage and highly diverse crop distribution [13].

The input features are constructed from Earth Observation data provided by the European Space Agency's **Sentinel-2** mission [4]. The study utilizes 805 Level-1C (L1C) Top-of-Atmosphere acquisitions [13]. While the 13 native multi-spectral bands possess varying spatial resolutions (10m, 20m, and 60m), the lower-resolution channels were spatially resampled to a consistent 10m resolution using a nearest neighbour approach [13]. In this process, the larger 20m and 60m pixels are subdivided to fit the 10m spatial grid, and each new 10m pixel is simply assigned the exact spectral reflectance value of its closest original parent pixel. This specific technique is utilized because it perfectly preserves the original, raw radiometric data without mathematically interpolating or inventing new intermediate values that could distort a crop's true spectral signature. To convert the raw digital numbers into a standardized Top-of-Atmosphere (TOA) reflectance scale, the pixel values were mathematically divided by the sensor's absolute quantification value (typically 10,000) extracted directly from the Sentinel-2 metadata. This ensures the neural networks process normalized, physically meaningful radiometric signals. No cloud screening was applied, maintaining the raw sequence with its natural atmospheric occlusions [13].

To facilitate multi-year sequence modeling, the dataset was preprocessed to isolate a strict temporal sequence across six consecutive agronomic years. An agronomic year in this context is defined as the continuous observation period from September 1st of the preceding year to September 1st of the harvest year. Consequently, the temporal sequence of satellite images acquired between September 1, 2016, and September 1, 2017, is strictly associated with the final 2017 harvest label. To construct the final dataset, strict filters were applied: fields were required to have a minimum size of 400 m^2 [13]. Crucially, an inward erosion buffer of 10 meters was applied to all parcel geometries before extracting the spectral signatures. Because Sentinel-2's highest spatial resolution is 10m per pixel, this erosion guarantees that only pure, unmixed pixels strictly residing within the field's interior are aggregated. Parcels whose valid geometric area fell below the 400 m^2 threshold after this buffering process were strictly excluded, naturally dropping fields that became effectively empty.

Furthermore, the dataset was filtered to retain only fields that maintained a consistent geometric presence across all six target years [13]. In agricultural administrative registries, parcels are frequently merged or subdivided by farmers between seasons; selecting only stable geometries ensures that the multi-year sequence models are analyzing the exact same plot of land over time without artificial

spectral shifts caused by changing borders. Additionally, the retained fields were filtered to include only those whose declared crop types possessed a dataset occurrence greater than 0.05% [13]. While this strict temporal footprint deliberately excludes intermittent fields, shifting borders, and extremely rare crops, it is mathematically necessary to establish a stable population for sequence evaluation. After filtering, the dataset encompasses **47 distinct crop classes** across approximately **300,000 crop fields** [13].

Crucially, rather than processing memory-intensive 2D spatial patches, each resulting crop field is collapsed into a single 1D entry representing the **spatial mean value** of the pixels belonging to that specific field on every acquisition date. Therefore, the unit of analysis and classification for all models in this study is the individual field. The data extraction and pre-processing pipeline utilized in this thesis was originally designed and implemented by Perantoni et al., allowing this research to bypass standard preprocessing bottlenecks and focus entirely on architectural development.

To rigorously evaluate the architectures without data leakage, the approximately 300,000 filtered fields were partitioned into Training (60%), Validation (20%), and Test (20%) splits. Rather than utilizing a naive random split, a stratified random sampling approach was employed based on the uniqueness of the multi-year label sequences. An agglomerative hierarchical clustering algorithm (utilizing average linkage) was applied to group agricultural fields that exhibited similar crop rotation patterns [13]. To account for identical crop rotations that are simply out of phase (e.g., a "Wheat-Maize-Soy" rotation started in 2016 versus 2017), a rotating Hamming distance metric was utilized, building upon the foundational sequence alignment logic established by Hamming [34]. This metric calculates the minimum distance across all possible circular shifts of the multi-year time series, effectively recognizing temporally shifted sequences as the same core agricultural strategy. The 60/20/20 split was subsequently executed across the resulting clusters. This prevents data leakage by ensuring the Test set evaluates the models on strictly unseen geographic parcels, while maintaining a proportionally representative distribution of the multi-year rotation patterns seen during training. This setup focuses the evaluation on the models' ability to generalize learned rotational rules to new geographical parcels and varying atmospheric noise, rather than testing their ability to predict entirely novel, unseen crop rotation strategies.

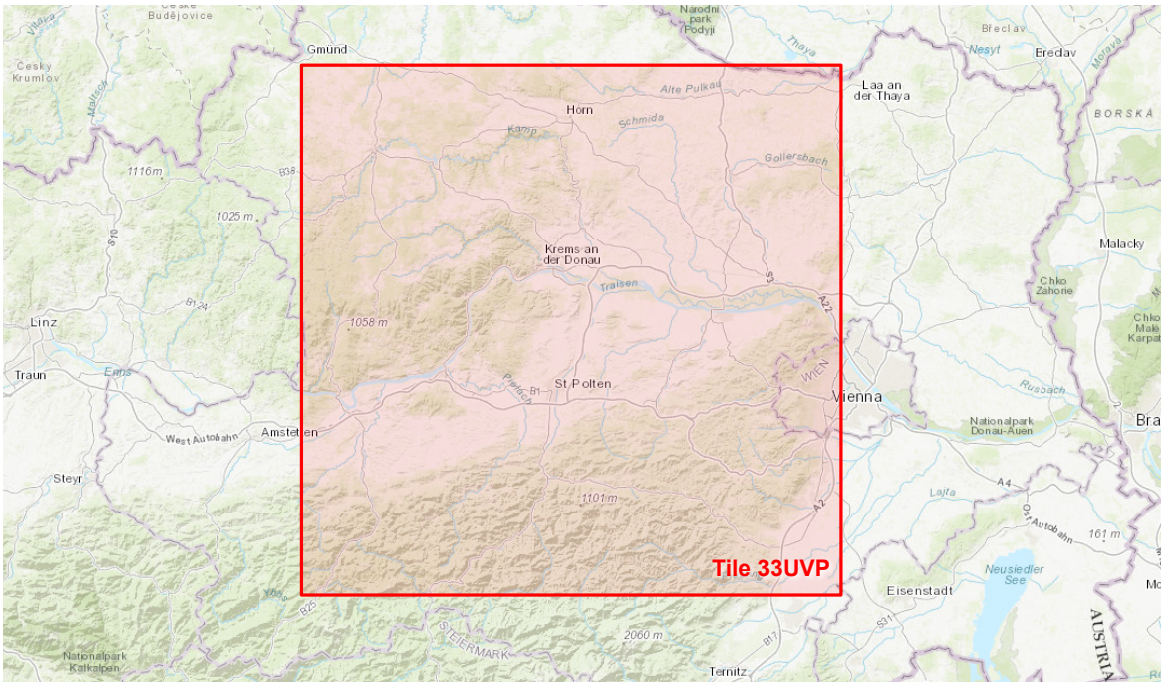


Figure 3.1: Geographic extent of Sentinel-2 MGRS Tile 33UVP over eastern Austria.

Geographically, the selected MGRS tile 33UVP covers a standard 100×100 km extent ($10,000$ km^2) over eastern Austria (encompassing regions such as Lower Austria and Vienna) as shown in Fig 3.1. This specific tile was selected because it guarantees consistent orbital coverage and contains a highly

diverse distribution of crops. Following the filtering processes, the dataset encompasses the 47 distinct crop classes detailed in Table 3.1.

3.1.2 HPC Environment and 1D Pixel-Based Extraction

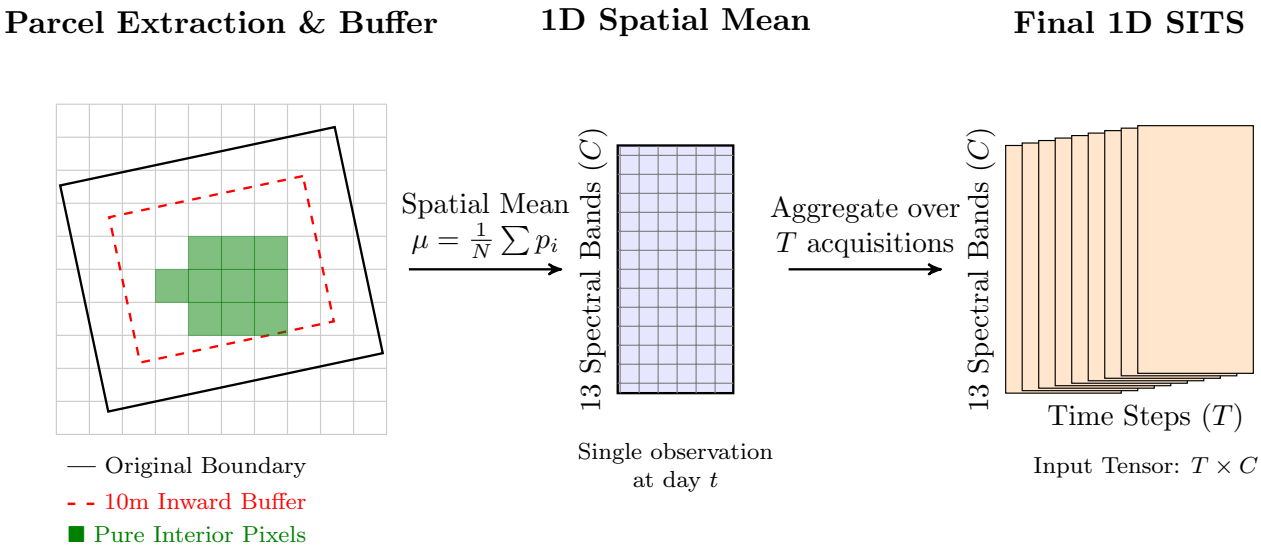


Figure 3.2: Detailed data preprocessing pipeline. To bypass I/O bottlenecks and mixed-pixel noise, field boundaries undergo a 10m inward erosion buffer. The remaining pure interior pixels are aggregated via a spatial mean into a 1D vector for each acquisition day, constructing the final $T \times C$ temporal sequence.

Training deep sequence models on massive, high-dimensional satellite time series is computationally demanding. To facilitate this, all experiments were executed on a dedicated High-Performance Computing (HPC) node equipped with an AMD Ryzen 7 5800X processor, 64GB of system RAM, and an NVIDIA GeForce RTX 3060 GPU, running on an Ubuntu 22.04 LTS environment.

A primary challenge in processing the INVEKOS dataset is the severe input/output (I/O) bottleneck caused by loading thousands of multi-year temporal sequences. Initial exploratory experiments attempted to utilize the full 2D spatial patches of the agricultural parcels. Because the massive dataset could not be stored on the local solid-state drive (SSD) and had to be hosted on a shared network folder, the immense volume of network-based disk reads caused severe GPU starvation. This rendered the 2D training process computationally infeasible within the available time constraints of this research. Consequently, this exploratory phase was aborted before a formal comparative benchmark could be established; exact performance metrics (such as peak VRAM, specific loading times, or precise per-epoch durations) for the 2D pipeline were not rigorously recorded.

To bypass this severe observed limitation, the experimental framework was forced to pivot to an optimized **1D pixel-based data pipeline** provided by Perantoni et al. [13]. Rather than processing broad spatial context, each sample is shaped as a 1D sequence containing the 13 spectral bands measured across the specific agricultural days of the year. This aggressive dimensionality reduction completely bypasses the network I/O limits, rendering the multi-year SITS training executable on a single computing node. Therefore, the utilization of the 1D pipeline is presented here not as a benchmarked computational contribution of this work, but rather as a strict engineering prerequisite to make the architectural comparisons possible.

3.1.3 Data Loading Strategies

To properly evaluate the impact of multi-year memory, the experimental framework utilized two distinct data loading strategies tailored to the architectural needs of the models.

For the single-year baseline models (1D-CNN and standalone Transformers), the dataset was structured using an **intra-year loader**. This loader decoupled the temporal sequence of the dataset, extracting and serving only a single growing season at a time. This constrained the baselines to rely

Table 3.1: Training set class distribution of the 47 crop classes, sorted by descending frequency.

Class ID	**Crop Name**	**Percentage (%)**
1	MOWING MEADOW/PASTURE THREE OR MORE USES	32.67
10	GRAIN MAIZE	7.31
2	MOWING MEADOW/PASTURE TWO USES	7.15
9	WINTER COMMON WHEAT	6.92
0	WINTER BARLEY	6.19
3	SILAGE MAIZE	5.64
13	DRESS	4.76
18	PERENNIAL PASTURE	4.57
4	WINTER TRITICALE	3.71
29	ALTERNATING MEADOW (EGART, ARABLE PASTURE)	3.14
26	SOYBEANS	2.36
24	WINTER RYE	1.99
7	SUMMER OATS	1.79
5	GREEN FALLOW	1.54
45	SUGAR BEETS	0.83
33	ALPINE FORAGE area	0.81
6	SINGLE-MOWN MEADOW	0.80
22	WINTER RAPESEED	0.78
21	SPRING BARLEY	0.75
15	CLOVER	0.66
16	MISCELLANEOUS FIELD FODDER	0.53
17	FORAGE GRASSES	0.50
28	CORN CORN COB MIX (CCM)	0.46
30	HUT WILLOW	0.43
11	WINTER MIXED CEREALS	0.41
8	BROAD BEANS (BROAD BEANS)	0.32
23	SUMMER MIXED CEREALS	0.26
27	WINTER SPELT (SPELZ)	0.26
38	LUCERNE	0.23
46	SEED MAIZE PROPAGATION	0.23
43	STARCHINDUSTRIAL POTATOES	0.23
12	TABLE POTATOES	0.22
20	GRASSLAND FALLOW	0.20
31	OIL PUMPKIN	0.17
35	PEAS - CEREALS MIXTURE	0.14
19	ELEPHANT GRASS (CHINESE REED, MISCANTHUS SINENSIS)	0.13
32	ENERGY WOOD WITHOUT ROBINIA	0.11
44	SEED POTATOES	0.11
25	SUMMER COMMON WHEAT	0.11
42	SINGLE-CULTURE FIELD VEGETABLES	0.10
40	SUMMER POPIES	0.09
37	GRAIN PEAS	0.08
14	STRAWBERRIES	0.07
39	WINTER CUMIN	0.07
41	SUNFLOWERS	0.07
34	TABLE APPLES	0.06
36	OTHER ARABLE LAND	0.05

entirely on localized, intra-year phenological signatures for classification.

For the multi-year hybrid architectures (Transformer-HMM and Transformer-LSTM), a **multi-year loader** was employed. This loader extracts continuous six-year SITS windows for each field, maintaining the strict chronological sequence across multiple growing seasons. This structured temporal context is what feeds the inter-year sequence models, allowing them to learn sequential agricultural practices.

3.1.4 Evaluation Metrics

Agricultural land cover datasets are inherently unbalanced; a few dominant crop types often account for the vast majority of cultivated land, while specialized crops represent a tiny fraction of the dataset. Consequently, relying solely on overall accuracy can yield highly misleading evaluations. To provide a comprehensive assessment, the models were evaluated using two primary metrics: **Overall Accuracy** and the **Macro F1-Score**.

Overall Accuracy (OA) measures the total proportion of correctly classified agricultural parcels across all available classes. It is defined as:

$$\text{OA} = \frac{\sum_{i=1}^{C} TP_i}{N}$$

where C is the total number of target classes (47 in this study), TP_i represents the True Positives for class i, and N is the total number of samples.

Macro F1-Score is the primary metric used to evaluate true model performance. The F1-score for a single class is the harmonic mean of Precision and Recall. To compute the Macro F1-score, the F1-score is calculated independently for each of the 47 classes and then averaged with equal weight across classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} \left(2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \right)$$

Unlike a weighted average, the Macro F1-score treats every class equally, regardless of its support size. A model can only achieve a high Macro F1-score if it successfully identifies both the dominant and the minority crop classes, making it a rigorous benchmark for this highly imbalanced dataset. To guarantee a strictly fair comparative analysis and validate the aggregate model behaviors, the Overall Accuracy and Macro F1-score are calculated using the exact same mathematical procedure, across the exact same field-year samples from the test set, and strictly over the identical 47 classes for all evaluated models (baselines, Transformer-LSTM, and Transformer-HMM).

3.2 Infrastructure and MLOps

The complexity of simultaneously training intra-year feature extractors and inter-year sequence heads required a robust software infrastructure to manage the workflow efficiently on the available hardware.

3.2.1 Training Framework and Optimization

The entire experimental codebase was developed in Python utilizing the **PyTorch** deep learning framework, wrapped within **PyTorch Lightning** [35]. PyTorch Lightning was selected to streamline the development process and ensure code reproducibility. While the framework natively supports distributed multi-node and multi-GPU training, all experiments in this thesis were executed concurrently as independent trials on a single NVIDIA GeForce RTX 3060 GPU.

The networks were trained using the **AdamW optimizer** [36], which explicitly decouples weight decay from the gradient updates to improve generalization. A crucial optimization strategy for the Transformer-HMM was the implementation of decoupled learning rates. By assigning independent learning rates to the neural backbone and the sequence head, the automated tuner could independently scale the weight updates, ensuring the neural network learned effectively without destabilizing the HMM's probabilistic transition matrices. All models were optimized against a Categorical Cross-Entropy (CCE) loss function.

3.2.2 Automated Hyperparameter Tuning

To ensure all baseline and hybrid models were compared fairly, an automated Machine Learning Operations (MLOps) pipeline was utilized. **Ray Tune** [37], integrated with the **Optuna** search algorithm [38], was deployed to conduct hyperparameter sweeps across concurrent trials on the same machine. This combination utilized Bayesian optimization to intelligently search the parameter space (specifically learning rates), aggressively pruning unpromising configurations. Furthermore, all model executions, configurations, and resulting performance metrics were systematically tracked and versioned using a centralized **MLflow** server [39].

To ensure rigorous and reproducible optimization, all hybrid architectures were tuned using an automated Ray Tune and Optuna MLOps pipeline. During the tuning phase, models were evaluated across 30 total trial configurations (tune_epochs = 5) to search over hyperparameters such as learning rates, weight decays, and LSTM dimensions depending on the model selected. The optimal configuration for each model was selected based on maximizing the validation Macro F1-score (`sval_f1_te`) on the validation set. For final reporting, the selected optimal configurations were trained for 40 epochs (final_epochs = 40) using a batch size of 32 and optimized with the AdamW optimizer. To account for weight initialization variance and ensure statistical robustness, all final test metrics reported in this study represent the mean and standard deviation computed across 3 independent training runs ($N = 3$ using random seeds 42, 43, and 44) using the optimal configurations.

To provide a concise overview of the selection process and the distinct configurations utilized across the model variants, a synthesis of the hyperparameters is presented in Table 3.2. Furthermore, the tuning search space was strictly tailored to the sequence head: HMM configurations decoupled the learning rate of the backbone from the transition matrix to ensure stability, while the LSTM configurations explored varying depths and hidden dimensions.

Table 3.2: Synthesis of the MLOps evaluation protocol, fixed architectural configurations, and model-specific tuning search spaces.

Protocol Element / Hyperparameter	**Configuration / Search Space**
Global Training Protocol	
Model Selection Criterion	Maximum Validation Macro F1-Score (`sval_f1_te`)
Tuning Strategy	Ray Tune & Optuna (30 total trials, 15 random startup trials)
Epochs	5 Epochs (Tuning phase) / 40 Epochs (Final test phase)
Statistical Robustness (Test Phase)	Mean and Std. Dev. computed across 3 seeds (42, 43, 44)
Fixed Architectural Parameters	
Batch Size	32
Transformer Architecture	6 layers, 8 heads, $d_{model} = 512$, $d_{inner} = 2048$
Transformer Dropout	0.4
Model-Specific Tuning Search Spaces (Log-Uniform & Choice)	
1D-CNN & Standalone Transformers	Backbone LR ($10^{-5} \rightarrow 10^{-2}$), Weight Decay ($10^{-6} \rightarrow 10^{-2}$)
HMM Models (Standalone & Hybrid)	Backbone LR ($10^{-5} \rightarrow 10^{-2}$), Matrix LR ($10^{-5} \rightarrow 10^{-2}$)
LSTM Models (Hybrid)	LR ($10^{-5} \rightarrow 10^{-2}$), Weight Decay ($10^{-6} \rightarrow 10^{-2}$)
	LSTM Hidden Dim (128, 256, 512), LSTM Layers (1, 2, 3)

3.3 Baseline Results (Intra-Year and Structural Controls)

The first phase of experimentation isolated the single-year models to quantify the value of non-linear feature extraction. To ensure statistical robustness and account for weight initialization variances, all baseline models were trained and evaluated across three distinct seeds.

Model	Test Accuracy (%)	Test Macro F1 (%)	Test Loss
Temporal 1D-CNN	55.30 ± 1.46	30.17 ± 0.50	1.43 ± 0.08
Discriminative Transformer	55.57 ± 0.59	33.18 ± 0.30	1.45 ± 0.01
Generative Transformer	53.47 ± 0.40	36.01 ± 0.51	1.54 ± 0.02
Linear Control (w/ HMM)	41.52 ± 0.28	05.23 ± 0.18	4.99 ± 0.06
Linear Control (w/ LSTM)	32.41 ± 0.26	01.04 ± 0.32	4.92 ± 0.01

Table 3.3: Performance comparison of the intra-year baselines and linear structural controls. Results are averaged across random seeds, displaying the mean and standard deviation.

3.3.1 Temporal 1D-CNN vs. Transformer Encoders

The **1D-CNN** established a strong baseline by effectively mapping local spectral variations. However, as shown in Table 3.3, it achieved a moderate overall accuracy of 55.30% but struggled significantly with class imbalance, recording a mean Macro F1-score of only 30.17%. The model's sliding receptive field of five observations likely struggled to bridge the temporal gaps caused by prolonged winter cloud cover.

Under the evaluated protocol, the discriminative Transformer achieved a mean Macro F1 of 33.18%, compared with 30.17% for the 1D-CNN. However, it is necessary to declare a structural limitation in this comparison: the Transformer models received explicit temporal information (the acquisition dates) via the positional encoding, whereas the 1D-CNN received no explicit time information and relied entirely on the implicit order of the irregular sequence. Consequently, this baseline experiment evaluates two systems with different information baselines, isolating the combined effect of self-attention and temporal encoding rather than self-attention alone.

Furthermore, the explicit comparison between the two Transformer formulations yielded critical insights. While the **Discriminative Variant** achieved a marginally higher overall accuracy (55.57% vs 53.47%), the standalone **Generative Variant** yielded superior performance on the minority classes, achieving a higher Macro F1-score (36.01% vs 33.18%). By removing the bias terms and enforcing a strict, normalized probability space, the Generative Transformer evaluates the pure spectral evidence without defaulting to dataset priors. Furthermore, the application of a temperature scalar ($\tau = 0.1$) acts as a critical hyperparameter to calibrate the entropy of the output distribution. Rather than producing perfectly uniform noise or collapsing into completely one-hot vectors (which would destroy and bypass the HMM's mathematical smoothing capabilities), this temperature calibration balances the emission likelihoods. It ensures they provide soft, meaningful probabilistic evidence to the downstream transition matrices [13].

3.3.2 The Necessity of Non-Linearity: Linear Controls

To validate the architectural necessity of the deep neural backbones, the multi-year sequence heads were deliberately paired with a **zero-memory linear backbone**. In these control configurations, the raw 1D SITS data was processed through a single linear projection before being passed to either the HMM or the LSTM.

The evaluated Linear-HMM configuration obtained a Macro F1 of 5.23%. This result motivates investigating the role of the feature extractor, but does not establish a general requirement for deep nonlinear backbones. The **Linear-HMM** configuration managed to maintain a baseline level of predictive power (5.23% F1-score and 41.52% Accuracy). Because the HMM relies on an explicit, mathematically rigid transition matrix, this agronomic "safety net" allowed it to partially smooth over the highly overlapping, low-confidence emission probabilities generated by the linear layer.

In contrast, the **Linear-LSTM** configuration struggled significantly, yielding a Macro F1-score of merely 1.04%. A deep inspection of the per-class metrics and confusion matrix revealed that this near-zero F1-score is the direct result of a severe predictive mode collapse, precisely explaining the highly skewed aggregate metrics. The Linear-LSTM completely abandoned the minority classes, predicting only 11 out of the 47 available classes. As demonstrated in Table 3.4, it disproportionately

collapsed its predictions into a few classes, severely over-predicting classes such as 1 and 15, while completely ignoring highly frequent classes like Class 2 (which had over 22,000 real occurrences but zero predictions).

Table 3.4: Per-class metric extraction for the Linear-LSTM demonstrating severe predictive mode collapse. The model abandons the long tail, predicting only 11 out of 47 classes.

Class ID	Real Freq. (Support)	Predicted Freq.	Precision	Recall	F1-Score
1	117,521	185,549	0.324	0.512	0.397
15	6,500	108,907	0.059	0.988	0.111
13	12,305	32,475	0.120	0.316	0.173
2	22,714	0	0.000	0.000	0.000
9	21,795	0	0.000	0.000	0.000

Lacking the predefined agronomic matrices of the HMM, the LSTM attempts to learn the complex, non-linear rules of crop rotation entirely end-to-end. However, because the linear projection was mathematically incapable of untangling the spectral signatures of 47 overlapping crop classes, the LSTM was fed near-random temporal features, preventing its recurrent gates from optimizing and causing the network to default to a majority-class guessing strategy.

3.4 Multi-Year Architecture Results

The final phase confronted the established **Transformer-HMM** Bayesian cascade against the novel **Hierarchical Transformer-LSTM**. For both multi-year architectures, the generative and discriminative variants of the Transformer backbone were evaluated to determine how the sequence heads best consume intra-year embeddings. Both architectures vastly outperformed the single-year baselines, definitively proving that analyzing multi-year historical crop rotations provides a massive predictive advantage.

Model	Test Accuracy (%)	Test Macro F1 (%)	Test Loss
Dis Tran-HMM	73.05 ± 0.55	63.15 ± 0.75	1.85 ± 0.05
Gen Tran-HMM (SOTA)	73.73 ± 0.50	63.95 ± 0.88	1.82 ± 0.04
Dis Tran-LSTM	74.01 ± 0.53	39.28 ± 0.65	2.34 ± 0.03
Gen Tran-LSTM (Novel)	**74.52 ± 0.48**	**40.12 ± 0.72**	**2.31 ± 0.04**

Table 3.5: Final performance comparison of the multi-year hybrid architectures across both backbone variants. Results are averaged across random seeds.

The **Transformer-HMM** provided excellent temporal smoothing, correcting isolated intra-year misclassifications by referencing its explicit, rigid transition matrices. To rigorously evaluate the backbone formulations, the Transformer-HMM was evaluated using three independent training runs with different random seeds on the same fixed train/validation/test split. Specifically, the dataset partitioning remained strictly identical across all runs to prevent evaluation variance, while the different random seeds solely dictated three stochastic components: the initial neural network weight distributions, the random dropout masks applied during training, and the shuffling sequence of the training dataloader. This methodology ensures the final averaged metrics are statistically robust against initialization randomicity, while evaluating exactly the same geographical parcels.

As hypothesized, the Generative Transformer-HMM (63.95% F1) outperformed its Discriminative counterpart (∼63.15% F1), validating that the HMM requires soft, unbiased emission probabilities. Overconfident predictions from the discriminative softmax effectively bypass the HMM's transition matrix, degrading its temporal smoothing capabilities. However, even at peak performance, the HMM's reliance on rigid, explicitly defined transition matrices (even if empirically initialized and optimized) slightly restricted its ability to adapt to anomalous or unconventional rotation strategies employed by

individual farmers.

In contrast, the **Hierarchical Transformer-LSTM** demonstrated highly competitive capabilities. Reaching a peak Macro F1-score of 40.12%, the LSTM proved capable of processing both backbone types, but importantly, the **Generative** variant yielded the optimal configuration (achieving 40.12% compared to the Discriminative's 39.28%).

This direct comparison provides a crucial architectural insight. Unlike the HMM, the LSTM does not mathematically require normalized probability distributions to function. However, the Discriminative Transformer directly outputs heavily biased posterior probabilities, effectively squashing the signal of minority crops early in the pipeline due to the dataset's extreme imbalance. By utilizing the Generative backbone, the intra-year extractor is forced into an unbiased prototypical space, preserving the weak spectral signatures of rare crops and passing a more balanced sequence of features to the LSTM. This allows the recurrent memory to better rescue minority classes, leading to its superior Macro F1.

While the Generative Transformer-LSTM marks a clear improvement over the standalone Generative Transformer's 36.01%, it ultimately fell short of the HMM's peak of 63.95%. To thoroughly explain this discrepancy between the high overall accuracy (74.52%) and the moderate Macro F1 (40.12%) of the Generative Transformer-LSTM compared to the HMM, a deep inspection of the per-class real versus predicted frequencies, precision, recall, and support was conducted across the Test set. The confusion matrices confirm that both the Generative Transformer-LSTM and the Generative Transformer-HMM successfully predict all 47 classes, entirely avoiding the mode collapse of the linear baseline. However, the performance gap stems directly from how the models handle the minority tail. The Generative Transformer-LSTM maintains extremely high precision and recall on the dataset's massive majority classes (e.g., achieving 0.886 F1-score on Class 1, and 0.979 F1-score on Class 2, mirroring its real support frequencies), which drives the Overall Accuracy to 74.52%. Yet, its recurrent gates seemed to struggle to maintain high precision and recall on the rarest minority crops when constrained by the unbiased generative prototypical space, dragging the unweighted Macro F1 down to 40.12%. In contrast, the HMM's explicitly initialized transition matrices provide a mathematically robust probabilistic prior that forcefully protects minority class transitions from being overwhelmed during training.

3.5 Discussion

The experimental results validate the central premise of this research: relying solely on intra-year phenology severely limits crop classification performance, particularly on highly imbalanced datasets.

3.5.1 Single-Year vs. Multi-Year Capabilities

The improvement from the single-year Transformers to the multi-year hybrids highlights a fundamental limitation of traditional models: they evaluate each growing season in a vacuum. In agricultural reality, planting decisions are heavily governed by strict multi-year crop rotation practices designed to prevent soil depletion. By feeding a sequence of yearly embeddings into the inter-year sequence heads (LSTM or HMM), the network learns these transitional probabilities. If the network is uncertain about a crop's spectral signature in year t, it can mathematically infer the likely crop by analyzing the confident predictions from years $t-1$ and $t+1$. The bidirectional models perform retrospective classification using observations from the full six-year sequence.

3.5.2 The Impact of Class Imbalance

The INVEKOS dataset suffers from severe long-tail class imbalance, dominated by staple crops. This imbalance is fully exposed by the baseline models. The 1D-CNN achieved an overall accuracy of 55.30%, but an F1-score of only 30.17%. This discrepancy indicates a network exhibiting high prediction frequencies for majority classes and significant error rates on minority crops [40].

The hybrid multi-year models significantly mitigate this failure mode. For instance, while the Generative Transformer-LSTM's overall accuracy reached 74.52%, its Macro F1-score jumped to 40.12%. This suggests that multi-year memory explicitly helps rescue minority classes. Rare crops are often planted as transitional phases within a longer crop rotation cycle. By recognizing the complete multi-year cycle, the LSTM predictions are probabilistically conditioned on the learned rotational sequence, preventing the model from defaulting to the statistically safe, majority-class prediction.

For instance, while the Generative Transformer-LSTM's overall accuracy reached 74.52%, its Macro

F1-score improved to 40.12%. Although this demonstrates that recurrent memory helps rescue minority classes compared to single-year models, it remained significantly lower than the HMM's 63.95%. This performance gap suggests that the LSTM's learned transition dynamics were more heavily influenced by class biases. We hypothesize that because the LSTM must learn multi-year transition rules entirely end-to-end from an extremely imbalanced dataset, its recurrent gates may inherently favor transitions toward majority staple crops to minimize its global loss. Conversely, the HMM leverages transition matrices that are explicitly initialized using empirical dataset frequencies. This initialization appears to provide a rigid probabilistic prior that protects minority class transitions from being overwhelmed during training, though targeted interpretability studies on the LSTM gating mechanisms are required to definitively confirm this internal behavior.

3.5.3 Computational Efficiency

Despite its architectural complexity, the Hierarchical Transformer-LSTM proved to be highly computationally efficient during training. This efficiency is attributed to the model's hierarchical design. Self-attention mechanisms allow for significantly more parallelization than processing deep temporal time steps sequentially through a recurrent network. By leveraging the Intra-Year Transformer to evaluate the dense intra-year sequences concurrently, the sequential computation burden on the inter-year LSTM is reduced to a sequence length of merely six steps (representing the six years of data). This demonstrates that fusing intra-year attention with inter-year recurrence is computationally feasible on standard workstation hardware, effectively bypassing the severe I/O bottlenecks associated with full 2D spatial patches, rather than proving absolute large-scale scalability.

4 Conclusions and Future Work

4.1 Conclusions

This thesis addressed the technical challenges of operationalizing deep learning for large-scale agricultural mapping, transitioning from single-year phenological analysis to multi-year crop rotation modeling. By shifting the INVEKOS Sentinel-2 dataset to an optimized 1D parcel-aggregated pipeline, the severe I/O constraints that typically hinder longitudinal SITS analysis were bypassed. This engineering optimization enabled the deployment of an automated MLOps pipeline leveraging PyTorch Lightning to fairly evaluate all architectural variants.

The systematic evaluation answered the primary research questions with quantitative evidence. Single-year baselines confirmed the limitations of analyzing growing seasons in isolation; the Temporal 1D-CNN and standalone Discriminative Transformer achieved limited Macro F1-scores of 30.17% and 33.18%, respectively. Furthermore, structural control tests explicitly motivated investigating the role of non-linear feature extractors: when the neural backbones were replaced with linear projections, the HMM cascade's performance dropped to a 5.23% Macro F1, demonstrating that raw SITS data must be cleanly untangled before transition modeling can occur.

Ultimately, the multi-year architectures significantly improved classification performance by leveraging historical crop rotation context. Both architectures successfully captured inter-year dependencies to improve over baseline models. The established Generative Transformer-HMM achieved an optimal Macro F1-score of 63.95%, while the novel Generative Transformer-LSTM reached 40.12%, significantly outperforming the 36.01% achieved by its single-year counterpart. Crucially, while the Transformer-LSTM demonstrated that recurrent memory cells can intrinsically model inter-year temporal dependencies end-to-end to boost accuracy, the results confirm that the explicitly initialized probabilistic transition matrices of the HMM remain the superior method for smoothing complex, highly imbalanced agricultural cycles. While the performance gap suggests differences in how transition dynamics are handled, further targeted interpretability studies are required to confirm the precise mechanisms driving this behavior.

4.2 Limitations and Future Work

While the optimized 1D pipeline and the Hierarchical Transformer-LSTM framework successfully mapped inter-year crop dynamics, several direct limitations emerged from this study, guiding the direction of future research.

4.2.1 Observed Limitations

- **Loss of Spatial Context:** The necessary shift to a 1D spatial-mean pipeline to resolve hardware bottlenecks completely discarded field-level spatial textures, boundaries, and neighborhood planting contexts.
- **Restricted Temporal Population:** The sequence evaluation required filtering the dataset to include only fields with a consistent geometric presence across all six years. This explicitly excluded intermittent fields, shifting borders, and the inherent declarative noise of administrative registries.
- **Information Baselines:** The explicit temporal distances injected into the Transformer models via the AgroDays Positional Encoding created an uneven information baseline when compared to the implicit sequential ordering of the 1D-CNN.

4.2.2 Future Work

Building upon these findings and acknowledging the aforementioned constraints, several promising avenues remain for advancing agricultural remote sensing:

- **Integration of State Space Models (SSMs):** Time and computational constraints limited the backbones evaluated in this thesis to convolutions and self-attention. Recent advancements in State Space Models, particularly the Mamba architecture, offer the potential to process extraordinarily long temporal sequences with linear scaling rather than the quadratic complexity of Transformers. Evaluating a Mamba-LSTM or Mamba-HMM cascade could further reduce training times and memory overhead.

- **Reintroducing Spatial Context:** With the temporal sequence modeling now optimized, future work should explore lightweight methods to reintroduce spatial context. Utilizing Graph Neural Networks (GNNs) or localized 2D super-pixels could allow the model to capture field boundaries and neighborhood planting patterns without incurring the massive computational cost of full 3D spatiotemporal convolutions.

- **Multimodal Data Fusion:** Optical Sentinel-2 time series are inherently irregular due to cloud cover, requiring interpolation or attention-based masking. Fusing these optical sequences with Synthetic Aperture Radar (SAR) data from Sentinel-1 would provide weather-independent observations to help mitigate cloud occlusions, though this would not guarantee a completely uninterrupted observation stream. Future iterations of the Hierarchical Transformer-LSTM could leverage parallel cross-attention mechanisms to seamlessly integrate these modalities.

- **Large-Scale Domain Adaptation:** The multi-year sequence rules learned by the recurrent models were intrinsically tied to the climatic and agronomic practices of the Austrian INVEKOS dataset. Investigating transfer learning techniques to adapt these pre-trained crop rotation memories to entirely different geographical regions or climates represents a critical next step toward global precision agriculture.

- **Recurrent Interpretability and Agronomic Validation:** While the Bidirectional LSTM successfully improved multi-year mapping accuracy, further analytical studies are required to determine exactly *what* the memory cells learned. Future work should implement targeted interpretability methods (such as analyzing the LSTM gating mechanisms or executing ablation studies on transition matrices) to verify whether the network extracts true agronomic rotation rules, or simply exploits label persistence and geographical redundancies.

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